Machine Learning

04 April 2024 15:48

Machine learning is a part of AI (Artificial Intelligence) where we make machines to learn & perform the tasks without human help.



① Training means learning data.

② Testing means predicting based on the data trained.



Ex:- Netflix analysis our past data, it will make make prediction of what next we would like to see.

Types of Machine Learning

① Supervised M.L → we provide labelled data (i/p & o/p columns) for training. This type is only used for making predictions.

I/p					o/p
Username	age	gender	Product id	Price	Returned or not
X		M	..	.	True
Y		F			False

② Unsupervised M.L → I/p column (unlabelled data) for training. It is not used prediction. It is used for grouping data. Eg: Customer segmentation.

③ Semi Supervised M.L → we have some data labelled & most of the data is unlabelled.

Twitter Sentiment analysis

#	comments	Sentiment (happy, sad, angry)	
200	..	..	I/p & o/p
20	..	X	I/p.
500	..		

First 200 comments (I/p & o/p) will be used to create Supervised M.L model. Use same model to make predictions of remaining 300 comments.

④ Reinforcement M.L → It is not going to learn any data. Here machine are called as agents & agents will learn based on the envt. For eg:- Autonomous car, Gaming AI, Robotics.

Supervised Machine Learning [Trained on i/p & o/p columns]

Supervised Machine Learning [Trained on i/p & o/p columns]

It is classified into 2 typey based on data type of target variable (o/p column)

- Regression
- Classification.

Regression – There are many regression algorithms which are used when target variable is of continuous (decimal value) data type.

Eg: House price data.

i/p					o/p [Target]
Sqft	Location	No. of floors	BHK	Rent	Price
			2		2.8
			2		1.2
					0.98
			2		
			3		3.5

List of Regression Algorithms under Supervised M.L:

1. Linear Regression.
2. Time series Forecasting
3. KNN Regression.
4. SVM Regression.
5. Decision Tree Regression.
6. XG BOOST
7. Gradient Descent Regression.
8. Random Forest Regression.

Classification → These algorithms are used when target variable is categorical (text data) or discrete (whole numbers).

Eg:- Diabetes Data set

i/p					o/p
Pname	age	Insulin	Glucose	BP	Result
X					Yes / 1 ← categorical discrete.
Y					No / 0
Z					No / 0

List of Classification algorithms under Supervised M.L

1. Logistic Regression.
2. KNN
3. SVM
4. Naive Bayes
5. Decision Tree classifier.
6. Random Forest classifier.
7. KG BOOST classifier.
8. Gradient Boosting.

③ SVM
④ Naive Bayes

⑦ KG Boost Classifier.
⑧ Gradient Boosting.
⑨ Bagging.

Regression / classification ?

① Flight Price Prediction → Regression
Target → Price → continuous

② Covid 19 Prediction → classification.
Target → Yes / No] Categorical.

③ Sales Prediction → Regression.
Sale → continuous.

④ Email classification → classification
Target → spam / Not spam] categorical.

⑤ Iris data → classification.
species → versicolor / setosa / virginica] Categorical.